

Elliot J. Voss

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EDUCATION:

University of Illinois Urbana-Champaign (UIUC)

BS, Computer Science, Data Science and Machine Learning Concentration

Urbana, IL

December 2024

PROFESSIONAL EXPERIENCE:

SimplyConvert

Machine Learning Engineer

Edwardsville, IL

May 2024 – August 2024

- Architected a robust AI case approval system that utilized machine learning algorithms to identify and block harmful prompts, enabling faster response times and improving case resolution efficiency by 60% across paralegal teams
- Built an application to automate housing location labeling in text, transforming data into JSON format and replacing manual data entry processes.
- Developed a comprehensive data management protocol integrating Amazon Kendra and S3 Buckets, resulting in the centralized storage of 3,000 critical legal documents that is now utilized by 40 legal firms for quicker access
- Programmed a sophisticated AI chatbot that allows users to specify large language models for summarizing information from military and legal documents, achieving a 40% reduction in average handling time for inquiries
- Implemented a comprehensive natural language processing tool to identify military units in 1,500 documents, improving identification accuracy by 30% and saving 50 hours of review time per month

SIUE Center for Predictive Analysis

Data Science Researcher

Edwardsville, IL

May 2023 – August 2023

- Developed and optimized predictive models for Gibberella ear rot contamination in corn, employing TensorFlow and Scikit-Learn to implement LASSO regression, Random Forest Classifiers, and Neural Networks
- Created and implemented a sophisticated Linear Mixed-Effects model in R to forecast Deoxynivalenol levels in crops, integrating weather data, genotype, and field variables
- Enhanced model performance through hyperparameter training, data standardization, principal component analysis, leading to a 20% reduction in error rates
- Streamlined data analysis using Microsoft Excel to filter and analyze data sets based on precipitation, temperature, and date ranges, which accelerated data processing time by 35%

University of Illinois

Software Engineering Project Lead

Urbana, IL

August 2023 – December 2023

- Led a team of 8 students to develop a dynamic weather map application using JavaScript, enhancing user experience through a responsive and intuitive interface
- Employed several agile development techniques to increase the productivity of team members by an average of 15%
- Engineered seamless API integration to access real-time location and weather data from over 40 cities
- Facilitated weekly cross-team meetings, driving collaboration and ensuring consistent progress towards project milestones, resulting in on-time delivery

EXTRACURRICULARS:

Vex Robotics

August 2021 – December 2024

- Collaborated with a 20-member team to secure a 14th place finish at the 2022 VEX World Championship, demonstrating strong teamwork and technical skills in robotics
- Developed real-time software to process sensor data, enabling the robot to execute dynamic and precise movements during competition, increasing reaction time and movement accuracy
- Engineered autonomous navigation algorithms, allowing the robot to avoid moving obstacles and perform complex tasks without human intervention
- Designed and implemented a Proportional-Integral-Derivative (PID) controller to optimize the robot's steering and speed, improving stability and performance in dynamic environments

TECHNICAL SKILLS:

- Machine Learning and Artificial Intelligence, Natural Language Processing
- C, C++, Python, Java, JavaScript, HTML, OCaml, x86 Assembly, Git, WebGL
- Database Design and Management, Data Mining, Data Analytics, SQL
- Databricks, Microsoft Azure, Power BI, Tableau, ETL Pipelines
- Embedded Systems, Robotics, Computer Vision, Real Time Operating Systems
- Systems Programming, Data Structures and Algorithms, Computer Architecture
- Dynamic Programming, Functional Programming, Object Oriented Languages
- Communication, Problem-Solving, Critical Thinking, Time Management, Collaboration, Leadership